1A3

(1)

1. A 9375 MHz uniform plane wave is propagating in polystyrene. If the amplitude of electric field intensity is 20 V/m and the material is assumed to be lossless, find
   a) β
   b) λ in polystyrene
   c) Velocity
   d) η
   e) Amplitude of magnetic field intensity

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$f = 9375\ MHz = 9.375 \times 10^9\ Hz$

$E_0 = 20\ V/m$

$\varepsilon_r = 2.56 \quad \mu_r = 1$

$\omega = 2\pi f = 2\pi(9.375 \times 10^9)$

$= 5.89 \times 10^{10}$

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12} \times 2.56}} = 1.874 \times 10^8 \ m/s$$

a) $$\beta = \frac{\omega}{v} = \frac{5.89 \times 10^{10}}{1.874 \times 10^8} = 314.301 \ rad/m$$

b) $$\lambda = \frac{v}{f} = \frac{1.874 \times 10^8}{9.375 \times 10^9} = 0.01999\ m \approx 1.999\ cm$$

c) $$H_0 = \frac{E_0}{\eta} \approx 0.085 \ A/m$$

c) $v = 1.874 \times 10^8\ m/s$

d) $$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{2.56}} = 235.625\ \Omega$$

(2)

An electromagnetic wave in free space has a wavelength of 0.20 m. When this same wave enters a perfect dielectric, the wavelength changes to 0.09 m. Assuming that $\mu_r = 1$, determine ϵ_r and the wave velocity in the dielectric.

$\mu_r = 1$

$\lambda_1 = 0.2\ m$

$\lambda_2 = 0.09\ m$

$v_1 = 3 \times 10^8\ m/s$

$v_2 = \frac{c}{\sqrt{\varepsilon_r}}$

$$f = \frac{c}{\lambda_1} = \frac{3 \times 10^8}{0.2} = 1.5 \times 10^9\ Hz$$

$$v = f\lambda_2 = (1.5 \times 10^9)(0.09)$$

$$= 1.35 \times 10^8\ m/s$$

$$v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$

$$\sqrt{\varepsilon_r} = \frac{c}{v} = \frac{3 \times 10^8}{1.35 \times 10^8}$$

$$= 2.222$$

$\varepsilon_r = 4.938$

3

The amplitude of **E** just inside a liquid is 10.0 V/m and the constants are $\mu_r = 1$, $\epsilon_r = 20$, and $\sigma = 0.50$ S/m. Determine the amplitude of **E** at a distance of 10 cm inside the medium for frequencies of (*a*) 5 MHz, (*b*) 50 MHz, and (*c*) 500 MHz.

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$E_0 = 10$ V/m

$\mu_r = 1 \quad \epsilon_r = 20 \quad \sigma = 0.50$ S/m

$d = 10\text{cm} = 0.1$ m

$$\alpha = \omega\sqrt{\frac{\mu_0\mu_r\epsilon_0\epsilon_r}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\epsilon}\right)^2}-1\right]}$$

a) $f = 5$ MHz

$\omega = 2\pi f = 2\pi \times 5\times10^6 = \pi \times 10^7$

$E(0.10) = 10e^{-(3.1415)(0.10)}$

$= 10e^{-0.31415}$

≈ 7.316 V/m

b) $f = 50$ MHz

$\omega = \pi \times 10^8$

$\alpha = 9.398$ Np/m

$E(0.10) = 10e^{-(0.10)(9.9345)}$

≈ 3.907 V/m

c) $f = 500$ MHz

$\omega = \pi \times 10^9$

$\alpha = 19.4511$ Np/m

$E(0.10) = 10e^{-(0.10)(19.454)}$

≈ 1.429 V/m

4

The electric field of a TV broadcast signal progagating in air is given by

$$\mathbf{E}(z,t) = 0.2\cos(\omega t - 6.5z)\mathbf{a}_x \text{ V/m}$$

(a) Determine the wave frequency ω and the wavelength λ.
(b) Sketch E_x as a function of t at $z = 0$ and $z = \lambda/2$.
(c) Find the corresponding $\mathbf{H}(z, t)$.

$E(z,t) = 0.2\cos(\omega t - 6.5z)\,a_x$ V/m

$\mu_r = 1, \quad \epsilon_r = 1$

$E_0 = 0.2, \quad \beta = 6.5$

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$E_0 = 0.2, \beta = 6.5$

a) $\lambda = \frac{2\pi}{\beta} = 0.9666\ m$

$\omega = V \times \beta = C \times \beta$

$= 1.95 \times 10^9\ rad/s$

$f = \frac{\omega}{2\pi} = 3.10 \times 10^8\ Hz = 310\ MHz$

b) i) $z = 0$

$E(z,t) = 0.2\cos(\omega t - 6.5z)\,a_z$ V/m

$E(0,t) = 0.2\cos(\omega t - 0)\,a_z$ V/m

ii) $z = \lambda/2$

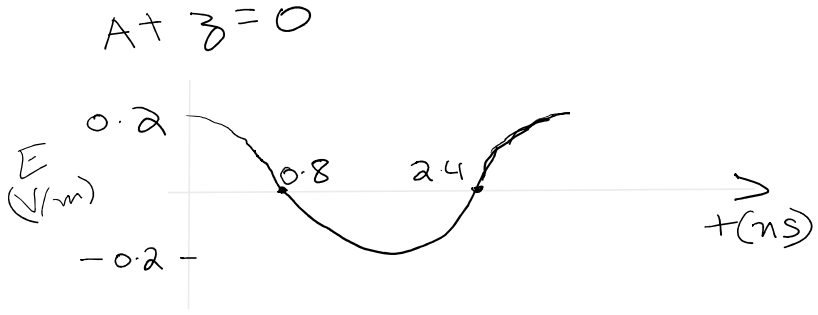
$\phi = \beta z = 6.5 \times \frac{\lambda}{2} = \frac{6.5 \times 0.9666}{2}$

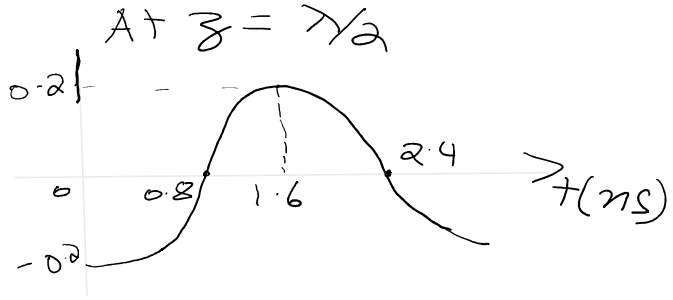
$= \pi$

$E(z,t) = 0.2\cos(\omega t - 6.5z)\,a_z$ V/m

$E\left(\frac{\lambda}{2},t\right) = 0.2\cos(\omega t - \pi)\,a_z$

$= -0.2\cos(\omega t)\,a_z$ V/m





c) $H_0 = \frac{E_0}{\eta_0} = \frac{0.2}{\sqrt{\frac{\mu_0}{\epsilon_0}}} = 0.53 \text{ mA/m}$

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$a_E \times a_H = a_{prop}$

$a_x \times a_H = a_z$

$a_H = a_y$

$H(z,t) = 5.3 \times 10^{-4} \cos(1.95 \times 10^9 t - 6.5z)\, a_y \text{ A/m}$

5

A manufacturer produces a ferrite material with $\mu = 750\mu_0$, $\varepsilon = 5\varepsilon_0$, and $\sigma = 10^{-6}$ S/m at 10 MHz.

(a) Would you classify the material as lossless, lossy, or conducting?
(b) Calculate β and λ.
(c) Determine the phase difference between two points separated by 2 m.
(d) Find the intrinsic impedance.

$\mu = 750\mu_0 \qquad \varepsilon = 5\varepsilon_0 \qquad \sigma = 10^{-6}$ S/m

$\mu_r = 750 \qquad \varepsilon_r = 5 \qquad f = 10$ MHz

$\omega = 2\pi f = 2\pi \times 10^7 \approx 6.283 \times 10^7$ rad/s

$\varepsilon = \varepsilon_r \varepsilon_0 = 5 \times 8.854 \times 10^{-12} \approx 4.427 \times 10^{-11}$ F/m

$\mu = \mu_r \mu_0 = 750 \times 4\pi \times 10^{-7} \approx 9.425 \times 10^{-4}$ H/m

a) $\tan\delta = \frac{\sigma}{\omega\varepsilon} = \frac{10^{-6}}{\omega \times \varepsilon} \approx 0.00036$

Since $\tan\delta << 1 \Rightarrow$ lossy dielectric

b) $\beta = \omega\sqrt{\mu\varepsilon} = 12.834$ rad/m

$\lambda = \frac{2\pi}{\beta} = 0.489$ m

c) $\Delta\phi = \beta \Delta x = 12.884 \times 2 = 25.668$ rad

d) $\eta = \sqrt{\frac{\mu}{\varepsilon}} = 4614.03\ \Omega$